



Cleared, Not Checked

An Audit of In-Room Fall-Alert Response Times Against Verified Welfare Checks on an Aged-Care Night Shift

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The problem

Aged-care providers grade night-shift response to fall-alert pages by how quickly a carer acknowledges them. Whether that speed reflects an actual walk-in check, or just the backlog of other pages, has gone untested.

Research questions

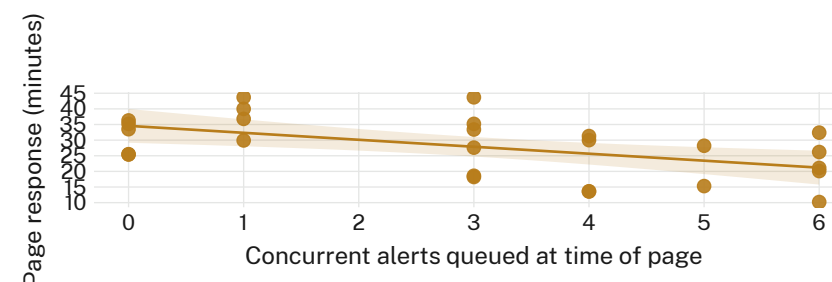
- Test whether response time predicts a verified in-room check
- Test whether queue depth explains it better than diligence does
- Compare the scorecard against an independently logged check record

Approach

- 26 night shifts · 3 residential facilities · 14 weeks; door-sensor entry as an independent check record
- Mixed-effects regression: response time against queue depth, shift hour, carer tenure
- Confirmed-fall pages (automatic emergency escalation) excluded

Findings

Response time tracked queue depth closely, but showed no comparable link to a verified check. It looked fastest on shifts least likely to end in a carer walking into the room.



Median page response fell from 34 to 14 minutes as concurrent queue depth rose from 0 to 6 alerts ($r = -0.52$, $n = 26$ shifts, 14 weeks); the same shifts' verified door-sensor entry rate showed no comparable trend ($r = 0.09$).

Implications

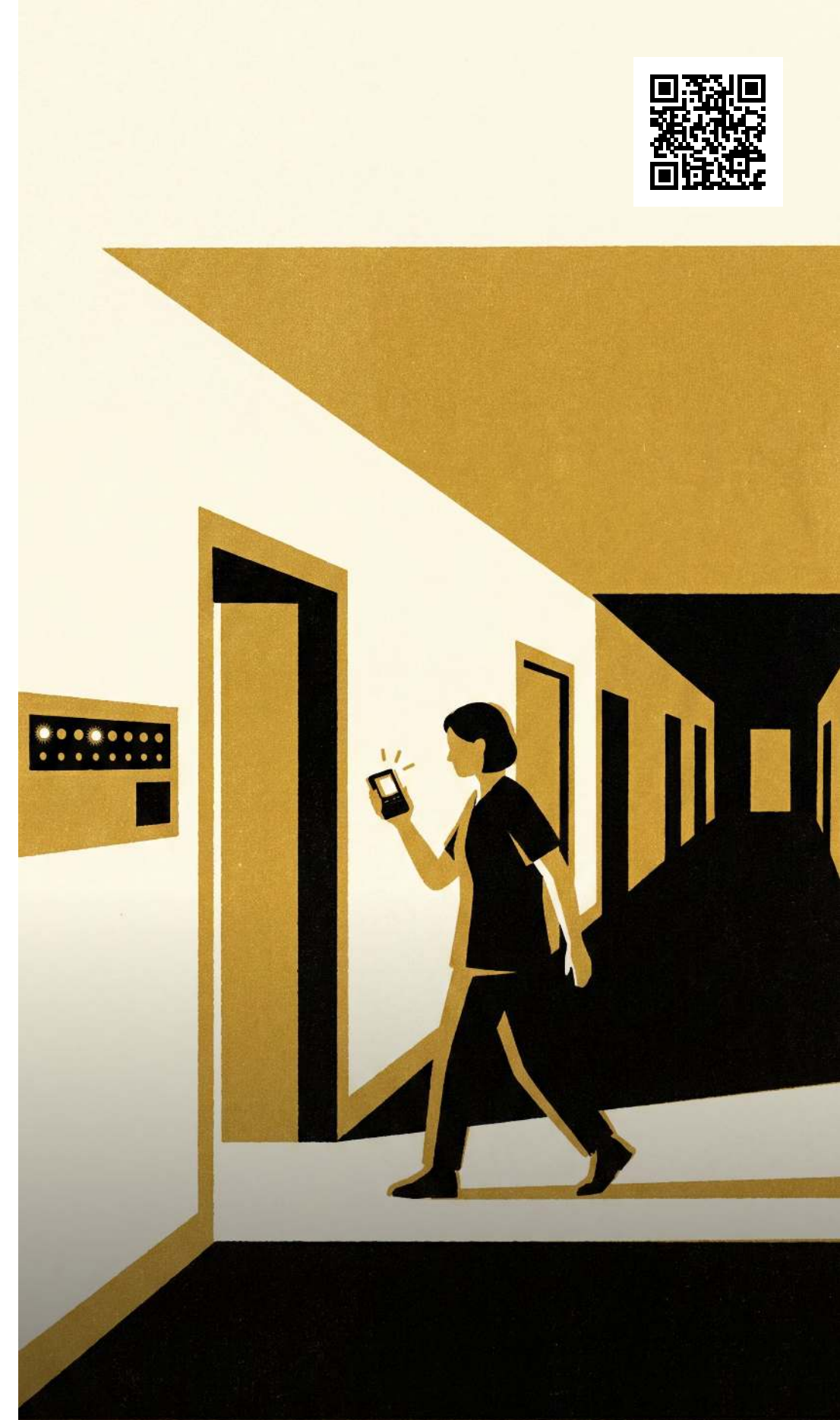
The scorecard rewards exactly the shortcut it was built to catch: clearing a page reads faster than walking to it. Next: validate against a wearable proximity log.

Selected references

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3. Igual, R., Medrano, C., & Plaza, I. (2013). Challenges, issues and trends in fall detection systems. *BioMedical Engineering OnLine*, 12, 66. doi:10.1186/1475-925X-12-66
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De-identified alert and door-sensor logs; no names retained; facility ethics approval 2025/213.

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“The page that clears fastest is the one nobody walked to.”

Median response fell 34 → 14 minutes as the queue grew, across 26 sampled night shifts.